



PHẠM PHƯỚC DŨNG (DANNY)

+84-932813052 | dung.phamphuoc308@gmail.com | www.linkedin.com/in/ppdung | Ho Chi Minh city, Vietnam
An experienced Software Engineer, holding distinguished bachelor's and master's degrees from Ho Chi Minh University of Technology (HCMUT). Displays extensive proficiency in POS (point-of-sale) systems and Service Robot utilization within restaurant environments. Shows a keen enthusiasm for developing software tailored to robotics and automotive applications.

Work Experiences

SMARTCAST Korea (smartcast.co.kr) - Seoul, South Korea - Remote

Feb 2025 - Present

Middle Software Engineer - C++/Qt/QML

- **Project:** DeepKDS – an AI-powered Kitchen Display System designed for seamless integration with POS systems worldwide
- **Contributions:**
 - Architected and developed DeepKDS from the ground up using **C++/Qt/QML** with multi-platform support (Windows, Android, macOS, Linux on Raspberry Pi), integrating **ZeroMQ** and **RabbitMQ** for real-time, low-latency data communication across distributed systems.
 - Collaborated to create DeepKDS AI Agent, a middleware leveraging **AI** to capture and interpret order data directly from printer output, eliminating dependency on POS vendor APIs and ensuring universal compatibility with POS systems worldwide.
 - Designed and implemented a secure data communication protocol between DeepKDS and the web server to enable registration, subscription management, and remote support in case of issues.
 - Designed a modern, intuitive UI/UX with AI-assisted insights, delivering a visually appealing and user-friendly interface aligned with current design standards.
 - Engineered a peer-to-peer remote control and screen-sharing subsystem built on **WebRTC** with hardware-accelerated H.264 encoding tailored to each platform (MediaCodec, Media Foundation, V4L2, x264), enabling support engineers to diagnose and resolve on-site issues remotely without dispatching technicians.
 - Built a fault-tolerant messaging layer featuring dual-broker failover, exponential-backoff reconnection, and transaction-level deduplication, ensuring kitchen displays continue serving orders through broker outages and the unstable network conditions typical of restaurant environments.
 - Implemented comprehensive internationalization across 10 locales with automatic menu translation, allowing a single build to serve restaurants in different countries without maintaining localized forks.
 - Integrated peripheral hardware and third-party services, including serial customer-pager protocols, thermal receipt printers, text-to-speech order announcements, and Kakao AlimTalk messaging for automated customer notifications.
 - Optimized on-device data persistence using **SQLite** with write-ahead logging, indexed queries, and automated retention policies, sustaining responsive performance on low-power embedded hardware across long-running deployments.
 - Instrumented the application with structured, category-based logging and symbol-resolved crash reporting, enabling rapid root-cause analysis of field issues across heterogeneous deployment environments.
 - Established a cross-platform **CI/CD** pipeline producing signed, self-contained builds for Windows, Android, and Raspberry Pi, shortening release turnaround and eliminating manual packaging errors.

SMARTCAST Korea (smartcast.co.kr) - Seoul, South Korea - Remote

Mar 2025 - Present

Middle Software Engineer - C++/Qt/QML

- **Project:** DeepDID – a customer-facing order-status display (Digital Information Display) that pairs with DeepKDS on the same restaurant network, showing pickup-counter callouts driven by the kitchen's live order feed.
- **Contributions:**
 - Architected and developed DeepDID from the ground up using **C++/Qt/QML** with multi-platform support (Windows, Android, Linux on Raspberry Pi), integrating **ZeroMQ** for peer-to-peer connectivity to Kitchen Display Systems on the local network and **RabbitMQ** for cloud command delivery, so a pickup-counter display keeps serving orders when either channel drops.
 - Designed and implemented a cloud-controlled configuration channel with tolerant JSON coercion for cross-version compatibility and a server-pushed version-snapshot rollback that copies the running binary aside before each install, letting operations remotely downgrade a device to a known-good previous release without dispatching a technician if a bad release is detected in the field.
 - Built a fault-tolerant messaging layer featuring dual-broker failover, exponential-backoff reconnection, and transaction-level deduplication, ensuring customer-facing displays continue announcing orders through broker outages and the unstable network conditions typical of restaurant environments.

- Engineered a peer-to-peer remote support subsystem built on **WebRTC** with hardware-accelerated H.264 encoding tailored to each platform (MediaCodec on Android, Media Foundation and D3D11 on Windows, V4L2 on Raspberry Pi), enabling support engineers to diagnose and resolve on-site display issues remotely without dispatching technicians.
- Designed a default-off master-switch mechanism for every opt-in remote-support capability (screen access, file transfer, clipboard sync), persisted in SQLite and controllable both from a device-side admin panel and from the cloud settings channel, so a store cannot be remotely observed until an operator explicitly consents.
- Built a hybrid **text-to-speech** announcer with two interchangeable engines — the platform speech API on hosts that provide one and a subprocess pipeline to a neural TTS binary on stock Raspberry Pi images that do not — delivering natural spoken order callouts on hardware that ships without a working system audio stack.
- Implemented the customer-facing rendering layer as an interchangeable-theme system with thirty-eight QML themes, ten locale translations, a video-and-image advertisement rotator, and a server-driven hot-swap between horizontal and vertical kiosk layouts, allowing a single build to serve store brands with radically different screen orientations and languages.
- Optimized on-device configuration persistence around **SQLite** with upsert semantics and an idempotent, marker-gated migration path from the legacy INI store, sustaining responsive settings changes on low-power embedded hardware while keeping upgrade safety through progressive schema evolution.
- Delivered Google Play compliance for the Android bundle by raising the target SDK to the current requirement and rebuilding both the app binary and its bundled OpenSSL shared libraries with 16 KB page-size alignment for arm64, verifying the segment alignment before every release so the app continues to install on devices running the new page size.
- Instrumented the application with rotating daily log files, rate-limited heartbeat logging so long-running kiosks do not fill their storage, a boot-time version marker so field-gathered logs can be attributed to a specific build, and a shared-memory single-instance guard preventing two GUI processes from fighting over the framebuffer on a kiosk.
- Established a cross-platform **CI/CD** pipeline on self-hosted Gitea Actions producing per-platform builds for Windows, Android, and Raspberry Pi, gated on a tag-to-source version match and auto-detecting the Qt, MinGW, and Android NDK layouts across runner install roots, shortening release turnaround and eliminating manual packaging errors.

SMARTCAST Korea (smartcast.co.kr) - Seoul, South Korea - Remote

Aug 2024 - Present

Middle Software Engineer - C++/MFC

- **Project:** Smartcast Agent - Middleware for Data Synchronization Between Local POS Systems and Kitchen Platform
- **Contributions:**
 - Took over and continued development of the software from the previous team leader, adding integration with additional local POS systems using API documentation provided by the POS vendors.
 - Designed and implemented a new user interface and optimized data flow chart, improving system usability and aligning the software with modern UX/UI standards.
 - Developed and optimized new features for Smartcast Agent, improving communication efficiency and reliability with the Smartcast server.

SMARTCAST Korea (smartcast.co.kr) - Seoul, South Korea - Hybrid

May 2023 - Aug 2024

Middle Software Engineer - C++/Qt/QML

- **Project:** OrderEAT - a tablet-based ordering application for restaurants
- **Contributions:**
 - Designed and implemented the OrderEAT from inception for both Windows and Android platforms, integrating communication with AWS servers for menu and order updates, Korean credit card payment processing, and interfacing with local POS and Kitchen Display Systems (KDS) in restaurants, utilizing **Qt/QML** technology.
 - Developed new functions for OrderEAT based on requirements from Korean team leader, utilizing **Qt/QML** for efficient implementation.
 - Managed task requirements through **Notion**, incorporated designs from **Figma**, and coordinated source code control via remote **GitHub** repositories.
 - Demonstrated adaptability and cultural fluency through two on-site assignments in Korea totaling four months, collaborating effectively with local teams.
 - Externally, spearheaded the development of a bespoke vending machine control software solution from scratch. This encompassed intricate communication protocols with embedded devices, integration with **AWS** for real-time sales updates, and the implementation of Korean credit card payment processing.

Team Leader

- **Project:** Autonomous Indoor Mobile Robot Integrated with Robotic Arm
- **Contributions:**
 - Spearheaded development planning and solved technical issues for an autonomous Indoor Mobile Robot product.
 - Streamlined project management using **JIRA** and maintained code repositories via **Bitbucket** and **GitHub**.
 - Pitched company products to investors at the Qualcomm® Vietnam Innovation Challenge 2022 Finale Event.
 - Led a team of three to Top 10 placement in the Qualcomm® Vietnam Innovation Challenge 2022, competing against leading start-ups.
 - Coordinated with cross-functional teams (hardware and embedded software) to successfully deliver the Mobile Robot product on schedule.
 - Researched and implemented autonomous indoor mobile robots with integrated robotic arms for advanced manipulation tasks.
 - Applied sensor fusion (Lidar, IMU, camera) to improve safety features and navigation reliability.
 - Developed autonomous navigation systems using **SLAM**, enabling efficient indoor mapping and localization.

Junior Software Engineer - C++/Qt/QML

- **Project:** Industrial Robot Controller for Delta Robot in pick-and-place applications
- **Team size:** 3
- **Contributions:**
 - Designed and implemented front-end GUIs for robotic controller and industrial vision applications on Windows and Linux using **Qt/QML**.
 - Developed back-end industrial vision software leveraging **AI** and **OpenCV** to detect and recognize objects on conveyor belts for automated pick-and-place systems.

Education Level

Masters Degree in Control Engineering & Automation, GPA 8.46/10.00

- Master thesis relating to develop indoor mobile robot using SLAM integrated robot arm for pick-and-place application.
- Operate and troubleshooting of AI based Bin picking automated system provided by Ajinextek, Joowon Technology, KETI ,ETRI and DGIST (Korea).

Bachelor Degree in Control Engineering & Automation, GPA 8.12/10.00

- Bachelor thesis relating to develop teaching-less application using computer vision for SCARA robot.
- Research on extracting depth map using Detectron2 and MiDaS.

Organisational Experience

- Ensure communication between the dormitory and students.

- Create strategies for sales and marketing to achieve high income.

Language

English	
• TOEIC 850	Jul 2022
• IELTS 6.5	2017